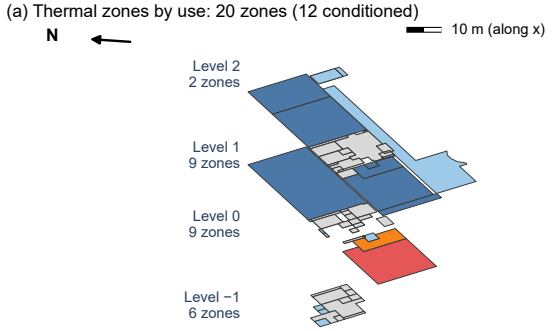
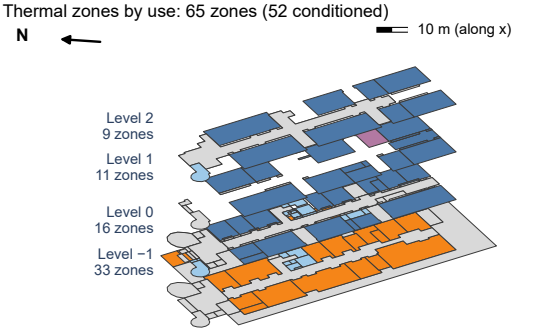
Building 10

(a) Thermal zones by use: 20 zones (12 conditioned)



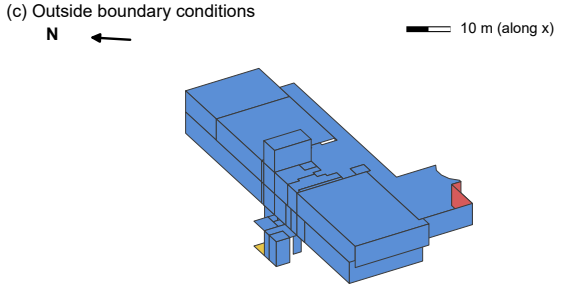
Building 09

(b) Thermal zones by use: 65 zones (52 conditioned)

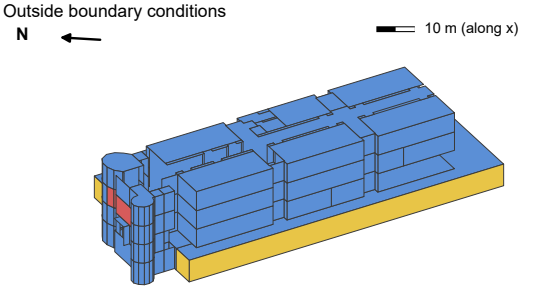


- Lecture hall / study room (B10: 4, B09: 20)
- Research laboratory (B10: 1, B09: 23)
- Office (B10: 0, B09: 1)
- Snack bar (B10: 1, B09: 0)
- Conditioned service / circulation (B10: 6, B09: 8)
- Unconditioned (stairs, shafts, storage, parking) (B10: 8, B09: 13)

(c) Outside boundary conditions

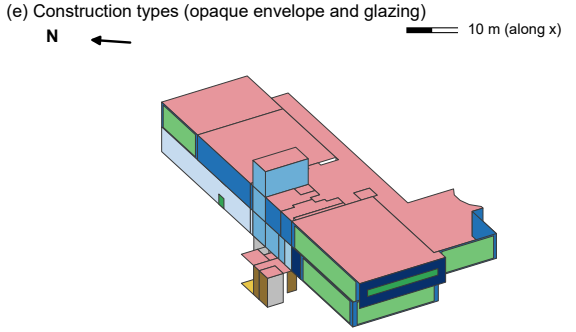


(d) Outside boundary conditions

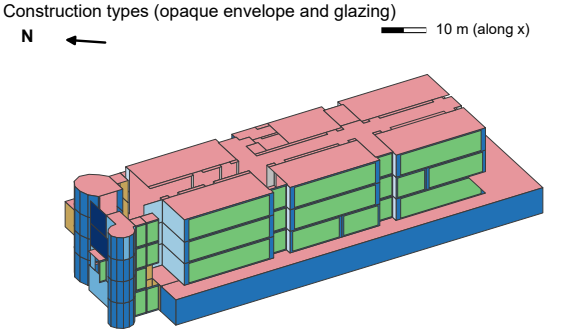


- Outdoors (B10: 135 surfaces, 6,049 m²; B09: 471 surfaces, 11,074 m²)
- Ground (B10: 15 surfaces, 197 m²; B09: 63 surfaces, 5,416 m²)
- Adiabatic (B10: 7 surfaces, 101 m²; B09: 5 surfaces, 96 m²)

(e) Construction types (opaque envelope and glazing)

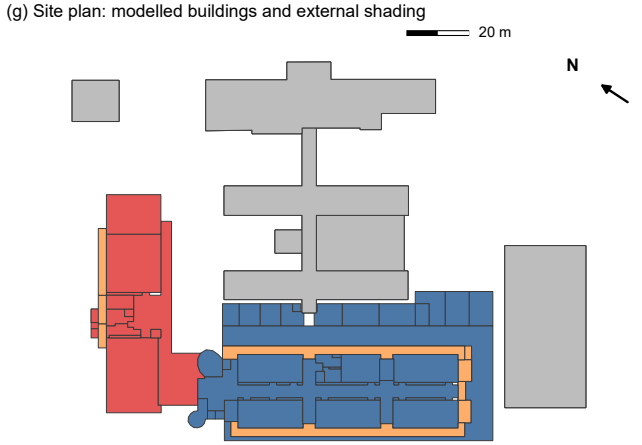


(f) Construction types (opaque envelope and glazing)



- External wall EW01: U 0.27 W/m²K (B10 118 m², B09 90 m²)
- External wall EW02: U 0.21 W/m²K (B10 1,574 m², B09 4,330 m²)
- External wall EW03: U 0.19 W/m²K (B10 218 m², B09 179 m²)
- External wall EW04: U 0.29 W/m²K (B10 12 m², B09 250 m²)
- External wall EW05: U 0.16 W/m²K (B10 164 m², B09 497 m²)
- Concrete wall FC300 (uninsulated): U 2.07 W/m²K (B10 190 m², B09 0 m²)
- Concrete wall FC300-2/-3 (insulated): U 0.20–0.45 W/m²K (B10 0 m², B09 1,136 m²)
- Partition types IW (exposed): U 0.40–0.73 W/m²K (B10 890 m², B09 2,097 m²)
- Roof: U 0.39 W/m²K (B10 1,590 m², B09 4,003 m²)
- Ground floor: U 0.30–0.52 W/m²K (B10 197 m², B09 4,003 m²)
- Floor over outdoor air: U 0.30 W/m²K (B10 1,393 m², B09 0 m²)
- Curtain-wall glazing: U 0.90 W/m²K (B10 1,090 m², B09 1,061 m²)
- Window glazing: U 0.90 W/m²K (B10 31 m², B09 4 m²)
- Curtain wall, HB default glazing: U 1.67 W/m²K (B10 0 m², B09 841 m²)

(g) Site plan: modelled buildings and external shading



- Building 10 (roofs)
- Building 09 (roofs)
- Balconies / overhangs (16 surfaces)
- Context buildings (64 surfaces)

Axonometric views: scale bar valid along the building x axis.
North arrows indicate true north (building north axis 45°).
U = thermal transmittance including surface films (EnergyPlus).
Zones spanning several levels are counted at each level.